实验三　静态路由的配置实验

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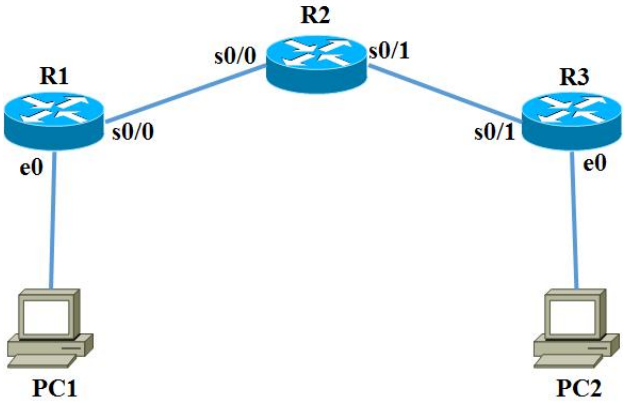
一、实验目的

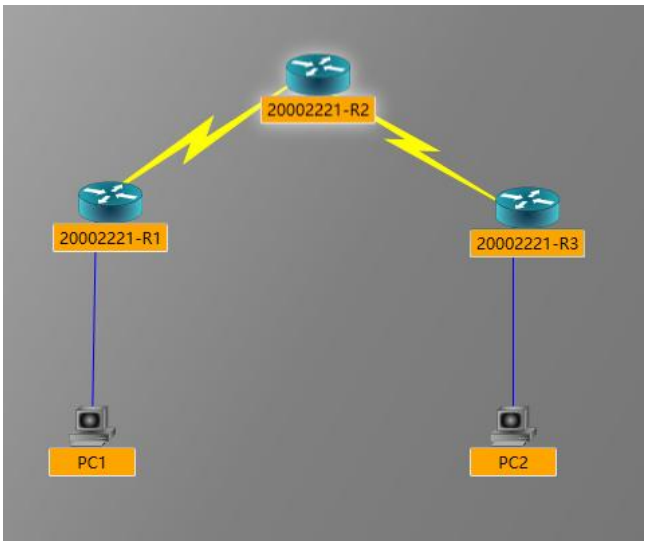
通过本次实验，让学生对路由器的设置方法和静态路由配置方式有一个初步的认识，对路由器各端口和手工建立路由表的方法有一个大概的了解。同时能够理解计算机网络中网络数据包的传递过程和路由器的转发机制。

二、实验装置与要求

三台模拟 Cisco2611 路由器，两台PC 机

按照下图的网络拓扑示意图，进行静态路由器的配置：





说明：路由器 R1、R2 和R3 使用Cisco 2611，其和PC1 和PC2 的接口的IP 地址分配如下表所

示。

设备	端口	IP	掩码	默认网关
R1	s0/0	192.168.2.1	255.255.255.0	
	e0	192.168.1.2	255.255.255.0	
R2	s0/0	192.168.2.2	255.255.255.0	
	S0/1	192.168.3.1	255.255.255.0	
R3	S0/1	192.168.3.2	255.255.255.0	
	e0	192.168.4.1	255.255.255.0	
PC1	e0	192.168.1.1	255.255.255.0	192.168.1.2
PC2	e0	192.168.4.2	255.255.255.0	192.168.4.1

三、实验内容

1. 实验步骤

配置 PC1 计算机的 IP 地址、子网掩码和网关（命令）：

```
C:>ipconfig /ip 192.168.1.1 255.255.255.0
C:>ipconfig /dg 192.168.1.2
```

配置 PC2 计算机的 IP 地址、子网掩码和网关：

```
C:>ipconfig /ip 192.168.4.2 255.255.255.0
C:>ipconfig /dg 192.168.4.1
```

截图显示配置路由器 R1 的步骤和命令：

(将路由器 R1 的名称定义为: 你的学号-R1, 例如: Router(config)#hostname 10190999-R1)

```
Router>en
Router#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.

Router(config)#hostname 20002221-R1
20002221-R1(config)#int e0
20002221-R1(config-if)#ip add 192.168.2.1 255.255.255.0
20002221-R1(config-if)#no shut
%LINK-3-UPDOWN: Interface Ethernet0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0, changed state to up
20002221-R1(config-if)#int s0/0
20002221-R1(config-if)#exit
20002221-R1(config)#int e0
20002221-R1(config-if)#no ip add 192.168.2.1 255.255.255.0
20002221-R1(config-if)#ip add 192.168.1.2 255.255.255.0
20002221-R1(config-if)#no shut
20002221-R1(config-if)#int s0/0
20002221-R1(config-if)#ip add 192.168.2.1 255.255.255.0
20002221-R1(config-if)#clock rate 64000
20002221-R1(config-if)#no shut
%LINK-3-UPDOWN: Interface Serial0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
%LINK-3-UPDOWN: Interface Serial0/0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to down
20002221-R1(config-if)#exit
20002221-R1(config)#ip route 192.168.3.0 255.255.255.0 192.168.2.2
20002221-R1(config)#ip route 192.168.4.0 255.255.255.0 192.168.2.2
20002221-R1(config)#end
20002221-R1#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
```

截图显示配置路由器 R2 的步骤和命令(路由器 R2 的名称定义为: 你的学号-R2)：

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname 20002221-R2
20002221-R2(config)#int s0/0
20002221-R2(config-if)#ip add 192.168.2.2 255.255.255.0
20002221-R2(config-if)#no shut
%LINK-3-UPDOWN: Interface Serial0/0, changed state to up
20002221-R2(config-if)#exit
20002221-R2(config)#int s0/1
20002221-R2(config-if)#ip add 192.168.3.1 255.255.255.0
20002221-R2(config-if)#clock rate 64000
20002221-R2(config-if)#no shut
%LINK-3-UPDOWN: Interface Serial0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to up
%LINK-3-UPDOWN: Interface Serial0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to down
20002221-R2(config-if)#exit
20002221-R2(config)#ip route 192.168.1.0 255.255.255.0 192.168.2.1
20002221-R2(config)#ip route 192.168.4.0 255.255.255.0 192.168.3.2
20002221-R2(config)#end
20002221-R2#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
```

截图显示配置路由器 R3 的步骤和命令(路由器 R3 的名称定义为: 你的学号-R3)：

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname 20002221-R3
20002221-R3(config)#int s0/1
20002221-R3(config-if)#ip add 192.168.3.2 255.255.255.0
20002221-R3(config-if)#no shut
%LINK-3-UPDOWN: Interface Serial0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to down
20002221-R3(config-if)#int e0
20002221-R3(config-if)#ip add 192.168.4.1 255.255.255.0
20002221-R3(config-if)#no shut
%LINK-3-UPDOWN: Interface Ethernet0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0, changed state to up
20002221-R3(config-if)#exit
20002221-R3(config)#ip route 192.168.2.0 255.255.255.0 192.168.3.1
20002221-R3(config)#ip route 192.168.1.0 255.255.255.0 192.168.3.1
20002221-R3(config)#end
20002221-R3#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
```

2．分别给出路由器 R1、R2、R3 路由表的屏幕截图。

```
20002221-R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default
       U - per-user static route

Gateway of last resort is not set

C    192.168.1.0 is directly connected, Ethernet0
S    192.168.3.0/24 [1/0] via 192.168.2.2
S    192.168.4.0/24 [1/0] via 192.168.2.2
C    192.168.2.0 is directly connected, Serial0/0
```

```
20002221-R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default
       U - per-user static route

Gateway of last resort is not set

S    192.168.1.0/24 [1/0] via 192.168.2.1
S    192.168.4.0/24 [1/0] via 192.168.3.2
C    192.168.3.0 is directly connected, Serial0/1
C    192.168.2.0 is directly connected, Serial0/0
```

```
20002221-R3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default
       U - per-user static route

Gateway of last resort is not set

C    192.168.4.0 is directly connected, Ethernet0
S    192.168.2.0/24 [1/0] via 192.168.3.1
S    192.168.1.0/24 [1/0] via 192.168.3.1
C    192.168.3.0 is directly connected, Serial0/1
```

3. 给出从一个 PC1 机 ping 另一个 PC2 机的屏幕截图。

```
C:>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:
Reply from 192.168.4.2: bytes=32 time=68ms TTL=241
Reply from 192.168.4.2: bytes=32 time=66ms TTL=241
Reply from 192.168.4.2: bytes=32 time=61ms TTL=241
Reply from 192.168.4.2: bytes=32 time=71ms TTL=241
Reply from 192.168.4.2: bytes=32 time=66ms TTL=241

Ping statistics for 192.168.4.2:
     Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
     Minimum = 61ms, Maximum =  71ms, Average =  66ms
```

四、思考题

1．路由器 r1 的 S0/0 端口与路由器 r2 的 S0/0 端口相连，当你首先配置好路由器 r1，然后配置路由器 r2，当你配置好 r2 的 S0/0 端口，输入 no shutdwon 命令后，你再次切换到路由器 r1，会看见 r1 命令窗口多显示出如下内容，解释为什么？

```
%LINK-3-UPDOWN: Interface Serial0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
10170999-r1#
```

答：配置 r2 的 s0/0 后输入 no shutdown，该端口会被激活，r1 的 s0/0 监测到状态变化后会重新启动激活端口。